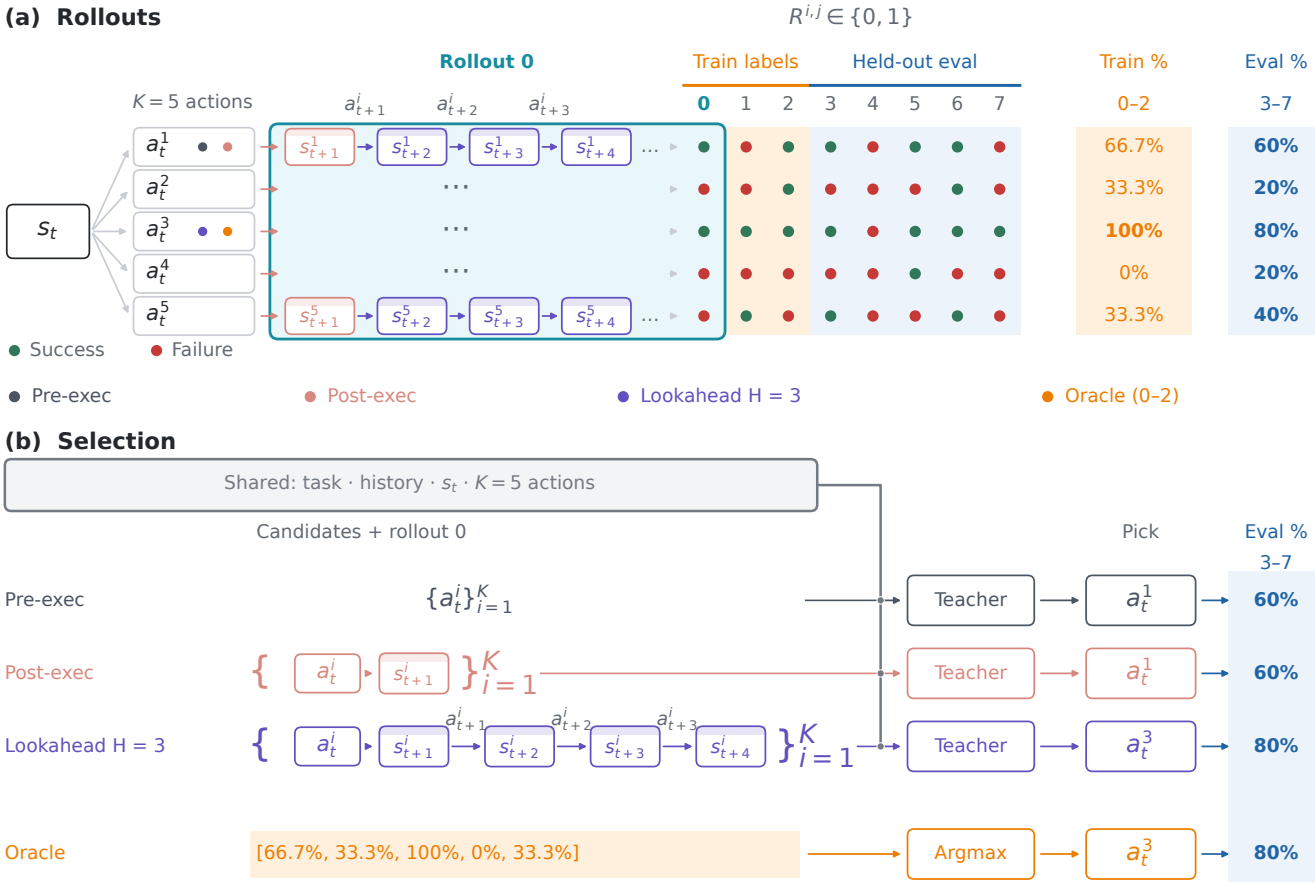
(a) Rollouts
$R^{i,j} \in \{0,1\}$
Rollout 0
Train labels
Held-out eval
Train %
Eval %
$K = 5$ actions
a^i_{t+1} a^i_{t+2} a^i_{t+3}
0 1 2 3 4 5 6 7
0–2
3–7
s_t
a^1_t s^1_{t+1} s^1_{t+2} s^1_{t+3} s^1_{t+4} … 66.7% 60%
a^2_t … 33.3% 20%
a^3_t … 100% 80%
a^4_t … 0% 20%
a^5_t s^5_{t+1} s^5_{t+2} s^5_{t+3} s^5_{t+4} … 33.3% 40%
Success
Failure
Pre-exec
Post-exec
Lookahead H = 3
Oracle (0–2)
(b) Selection
Shared: task · history · s_t · $K = 5$ actions
Candidates + rollout 0
Pick
Eval %
3–7
Pre-exec
$\{a^i_t\}^K_{i=1}$
Teacher
a^1_t
60%
Post-exec
$\{ a^i_t \to s^i_{t+1} \}^K_{i=1}$
Teacher
a^1_t
60%
Lookahead H = 3
$\{ a^i_t \to s^i_{t+1} \xrightarrow{a^i_{t+1}} s^i_{t+2} \xrightarrow{a^i_{t+2}} s^i_{t+3} \xrightarrow{a^i_{t+3}} s^i_{t+4} \}^K_{i=1}$
Teacher
a^3_t
80%
Oracle
[66.7%, 33.3%, 100%, 0%, 33.3%]
Argmax
a^3_t
80%